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Saturday, November 2, 2024 10:22 PM

4.5 Rates of Change in Trigonometric Functions

(1) Average Rate of Change

Average Rate of Change = $\dfrac{f(x_2)-f(x_1)}{x_2-x_1}$

$\dfrac{y_2-y_1}{x_2-x_1}$ slope of secant

(2) Instantaneous Rate of Change

Instantaneous Rate of Change= $\dfrac{f(a+h)-f(a)}{h}$, $h \to 0$ → slope of tangent

a k d c=0

Example #1: Consider the trigonometric function $y = -3\sin\left[\frac{1}{2}\left(x-\frac{\pi}{3}\right)\right]$.

mapping Rule : $(x, y) \to \left(2x+\frac{\pi}{3}, -3y\right)$

(a) Sketch one cycle of the function.

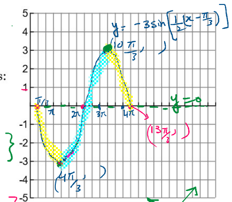
$(0, 0) \to \left(\frac{\pi}{3}, 0\right)$

$\left(\frac{\pi}{2}, 1\right) \to \left(\frac{4\pi}{3}, -3\right)$

$(\pi, 0) \to \left(\frac{7\pi}{3}, 0\right)$

$\left(\frac{3\pi}{2}, -1\right) \to \left(\frac{10\pi}{3}, 3\right)$

$(2\pi, 0) \to \left(\frac{13\pi}{3}, 0\right)$



(b) Determine an interval where the avg. rate of change is:

(i) positive (ii) negative (iii) zero

(i) $x \in \left[\frac{4\pi}{3}, \frac{10\pi}{3}\right]$

(ii) $x \in \left[\frac{\pi}{3}, \frac{4\pi}{3}\right]$ or $x \in \left[\frac{10\pi}{3}, \frac{13\pi}{3}\right]$

(iii) $x \in \left\{\frac{4\pi}{3}\right\}$ $x \in \left\{\frac{10\pi}{3}\right\}$

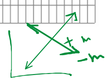
(c) Determine a point where the inst. rate of change is:

(i) positive (ii) negative (iii) zero

(i) $x = \frac{5\pi}{3}$

(ii) $x = \frac{2\pi}{3}$

(iii) $x \in \left\{\frac{4\pi}{3}\right\}$ $x \in \left\{\frac{10\pi}{3}\right\}$



(d) Calculate the average rate of change for $\frac{\pi}{2} \le x \le \frac{3\pi}{2}$.

$$AROC = \frac{f\left(\frac{3\pi}{2}\right) - f\left(\frac{\pi}{2}\right)}{\frac{3\pi}{2} - \frac{\pi}{2}}$$

$$= \frac{-2.898 + 0.776}{\pi}$$

$$\boxed{AROC \doteq -0.675}$$

$y = -3\sin\left[\frac{1}{2}\left(x-\frac{\pi}{3}\right)\right]$

$f\left(\frac{3\pi}{2}\right) = -3\sin\left[\frac{1}{2}\left(\frac{3\pi}{2}-\frac{\pi}{3}\right)\right]$

$= -3\sin\left(\frac{7\pi}{12}\right)$

$\boxed{f\left(\frac{3\pi}{2}\right) \doteq -2.898}$

$f\left(\frac{\pi}{2}\right) = -3\sin\left[\frac{1}{2}\left(\frac{\pi}{2}-\frac{\pi}{3}\right)\right]$

$= -3\sin\left(\frac{\pi}{12}\right)$

$\boxed{f\left(\frac{\pi}{2}\right) \doteq -0.776}$

(e) Describe how the instantaneous rate of change varies over the interval [0, 4π].

The IROC decreases, increases, decreases.
The IROC is zero at max and min

n .

Example #1: Consider the trigonometric function $y = -3\sin\left[\frac{1}{2}\left(x-\frac{\pi}{3}\right)\right] + 0$ Page - 37

$a = -3$

$|a| = 3$

$k = \frac{1}{2}$

$P = \frac{2\pi}{k} = \frac{2\pi}{1/2}$

$P = 4\pi$

$|\pi| \pi | \pi | \pi|$

$c = y = 0$ (No v.d)

Phase shift $= \frac{\pi}{3}$ (right)

max $= |a| + c = 3 + 0$

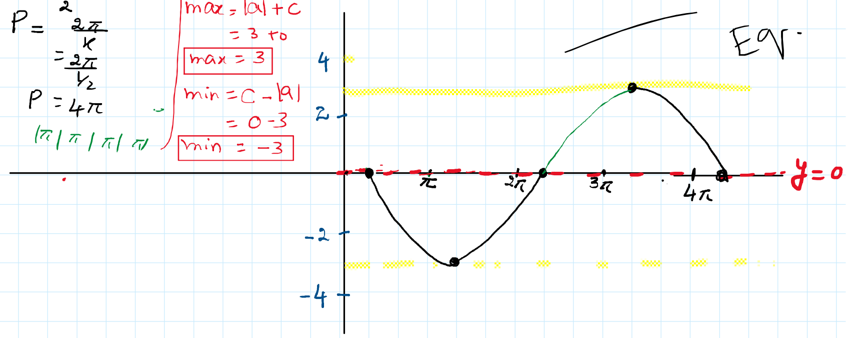
$\boxed{max = 3}$

min $= c - |a| = 0 - 3$

$\boxed{min = -3}$

reflected across x-axis

base function $y = \sin\theta$



Example #2: The position of a particle as it moves horizontally is described by the equation $s(t) = 12\sin\left(\frac{\pi t}{90}\right) + 15$, where s is the displacement, in metres, and t is the time, in seconds.

(a) Calculate the average rate of change of $s(t)$ for the following intervals:

(i) 5 s to 10 s

$$A.R.O.C = \frac{s(10) - s(5)}{10 - 5}$$

$$\doteq \frac{19.10424172 - 17.08377813}{5}$$

$$\doteq \frac{2.020463588}{5}$$

$$\doteq 0.404 \text{ m/s}$$

$$S(10) = 12\sin\left(\frac{10\pi}{90}\right) + 15$$

$$\doteq 19.10424172$$

$$S(5) = 12\sin\left(\frac{5\pi}{90}\right) + 15$$

$$\doteq 17.08377813$$

(ii) 9 s to 10 s

$$A.R.O.C = \frac{S(10) - S(9)}{10 - 9}$$

$$\doteq \frac{19.10424172 - 18.70820393}{1}$$

$$\doteq 0.396 \text{ m/s}$$

$$S(10) = 12\sin\left(\frac{10\pi}{90}\right) + 15$$

$$\doteq 19.10424172$$

$$S(9) = 12\sin\left(\frac{9\pi}{90}\right) + 15$$

$$\doteq 18.70820393$$

(b) Estimate the instantaneous rate of change of $s(t)$ at $t = 10$s.

$$I.R.O.C = \frac{f(10 + 0.001) - f(10)}{0.001}$$

$$= \frac{f(10.001) - f(10)}{0.001}$$

$$\doteq \frac{19.10463533 - 19.10424172}{0.001}$$

$$\doteq 0.394 \text{ m/s}$$

$$s(t) = 12\sin\left(\frac{\pi t}{90}\right) + 15$$

$$S(10.001) = 12\sin\left[\frac{10.001\pi}{90}\right] + 15$$

$$\doteq 19.10463533$$

$$S(10) = 12\sin\left(\frac{10\pi}{90}\right) + 15$$

$$\doteq 19.10424172$$

(c) What physical quantity does this instantaneous rate of change represent?

It represent the instantaneous velocity at 10 seconds